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B.Tech. V Semester

CS551PC: COMPUTER NETWORKS LAB

(Common to CSE, CSBS, CSE(AI&ML), CSE (Data Science))

Course Objectives:

- To understand the working principle of various communication protocols.
- To understand the network simulator environment and visualize a network topology and observe its performance
- To analyze the traffic flow and the contents of protocol frames

Course Outcomes:

- Implement data link layer farming methods
- Analyze error detection and error correction codes.
- Implement and analyze routing and congestion issues in network design.
- Implement Encoding and Decoding techniques used in presentation layer.
- To be able to work with different network tools.

List of Experiments:

1. Write a program to implement framing.
i. Fixed framing, ii. Variable framings
2. Write a program to implement stuffing techniques.
i. Bit-stuffing ii. Character stuffing
3. Write a program to implement checksum
4. Write a program to compute CRC code for the polynomial CRC-12
5. Develop a simple data link layer that performs the flow control using the sliding window protocol, and loss recovery using the Go-Back-N mechanism.
6. Write a program to implement classful addressing.
7. Write a program to implement classless addressing.
8. Write a program to implement Dijsktra's algorithm to compute the shortest path through a network
9. Write a program to divide a given network into n-sub networks using CISCO Packet Tracer.
10. Implement distance vector routing algorithm for obtaining routing tables at each node.
11. Write a program to implement Link state routing
12. Write a program for congestion control using Leaky bucket algorithm
13. Implement data encryption and data decryption.
14. Basic Trouble shooting commands ipconfig,ping ,tracert,nslookup,arp etc.

TEXT BOOK:

1. Computer Networks – Andrew S Tanenbaum, David.j.Wetherall, 5th Edition. Pearson Education/PHI,2011.